

Volume 35

Sentinel Query & Discovery Language (SQDL)

Status: Core Architecture

Priority: Critical

Vision

Project Sentinel should have its own security language.

Not SQL.

Not Elasticsearch syntax.

Not SIEM query languages.

A language designed specifically for cybersecurity.

Users should be able to ask questions naturally.

Examples:

Show Finance laptops

or

Find websites

or

Show all mobile devices

The platform translates this into object queries.

Why SQDL?

Current security platforms require users to learn:

- SQL
- KQL
- SPL
- Lucene
- Elasticsearch
- Vendor-specific languages

Instead, Project Sentinel provides **one consistent language** across every module.

Universal Query Model

Everything is queried through objects.

Objects

No matter whether the object is a:

- laptop,
- website,
- cloud account,
- API,
- employee,
- mobile device,

the syntax remains the same.

Query Object

Every query is stored.

Query ID

Queries themselves become reusable objects.

Saved Queries

Organizations create reusable searches.

Examples:

- Devices needing updates.
- Unsupported software.
- Expiring certificates.
- Unverified cloud resources.
- Open investigations.

One click.

Always available.

Relationship Queries

Because we have the Knowledge Graph.

Users can ask:

Show

The platform traverses relationships automatically.

Timeline Queries

Example

Show

or

Show

Everything is time-aware.

Trust Queries

Example

Find

Results are ranked by business importance.

Evidence Queries

Example

Show

Evidence freshness becomes searchable.

Mission Queries

Example

Show

Everything is connected.

Visual Query Builder

Many users will never type queries.

Instead:

Asset Type

Everything can be built visually.

Query Explanation

One of our biggest innovations.

Every query explains itself.

Example:

This query searches for laptops assigned to the Finance department whose latest verified Trust Score is below 95% and which have not been reassessed within the last 30 days.

This improves usability and transparency.

Query Optimization

The engine analyzes:

- Frequently used queries.
- Slow queries.
- Duplicate queries.
- Similar queries.

It recommends improvements.

Query Permissions

Not everyone can query everything.

Permissions apply to:

- Departments.
- Tenants.
- Regions.
- Classified assets.
- Personal information.
- Incident data.

Security applies to queries as well as objects.

NEW CORE ENGINE

Semantic Query Engine

Instead of requiring exact syntax.

The engine understands intent.

Example:

Computers

These can be treated as related concepts where appropriate, while preserving precise filtering when needed.

NEW CORE ENGINE

Query Recommendation Engine

The platform observes common usage patterns.

Example:

Users who review endpoint health every Monday may receive a suggestion to save that search as a reusable dashboard widget or scheduled report.

Recommendations remain optional and configurable.

NEW CORE ENGINE

Query Federation Engine

Organizations often operate across multiple environments.

The Federation Engine allows authorized queries across:

- Headquarters.
- Regional offices.

- Cloud environments.
- Subsidiaries.
- Managed tenants.

Results are merged into a single evidence-backed view.

NEW IDEA

Cyber Search

A universal search bar.

Instead of navigating menus.

Users type:

Finance VPN

or

Laptop John Smith

or

MFA rollout

or

Expired certificates

The platform searches:

- Objects.
- Missions.
- Evidence.
- Policies.
- Reports.
- Incidents.
- Knowledge Graph.

One search.

Everything.

NEW IDEA

Guided Search

Home users don't need a query language.

Instead, the platform asks:

What are you looking for?

- My computer
- My phone
- My Wi-Fi
- My website
- My accounts

Each answer launches an appropriate guided search.

NEW IDEA

Search as Navigation

Instead of menu-driven software.

Search becomes navigation.

This reduces the learning curve dramatically.

PS-ADR-0030

Search-First Platform

Status: Accepted

Decision

Project Sentinel shall adopt a search-first navigation model where users can locate objects, missions, evidence, policies, and reports through a unified discovery interface.

Rationale

Large cybersecurity platforms become difficult to navigate as they grow. A search-first model improves usability, reduces training requirements, and supports both novice and expert users.

NEW LONG-TERM VISION

Cyber Knowledge Explorer

I think we should go even further.

Imagine a visual explorer.

Organization

Users simply click relationships to move through the cyber environment.

No menus.

No deep navigation trees.
Just exploration.

NEW IDEA — One I believe could become a defining feature

Sentinel Workspace

Instead of separate pages, users work in **Workspaces**.

Examples:

- Executive Workspace.
- SOC Workspace.
- Incident Workspace.
- Bug Bounty Workspace (authorized targets only).
- Endpoint Workspace.
- Mobile Workspace.
- Cloud Workspace.
- Compliance Workspace.
- Home Workspace.

Each workspace combines:

- Search.
- Dashboards.
- Objects.
- Missions.
- Queries.
- Evidence.
- Reports.

This means every role sees exactly what they need without being overwhelmed by the rest of the platform.

One of the most important engineering principles so far

I want to formally add another design rule that reflects everything you've emphasized since the first day:

The platform must scale in complexity without scaling in user effort.

That means:

- A home user protecting one laptop should need only a few clicks to understand their security status.
- A small business should be able to manage dozens of devices without becoming overwhelmed.
- A global bank or government agency should be able to navigate hundreds of thousands of assets using the same consistent concepts: **Objects, Missions, Evidence, Knowledge, Trust, and Workspaces.**

If we preserve that principle throughout implementation, I believe Project Sentinel will stand out not because it has the most features, but because it makes advanced cybersecurity understandable at every scale.

Looking Ahead

The next foundational volume will define the **Sentinel AI Orchestration Framework**. Unlike many products that simply "add AI," we will design a framework where AI is modular, explainable, replaceable, privacy-aware, and always constrained by the evidence model and human oversight. That will allow organizations to use local models, cloud models, or future AI technologies without redesigning the rest of Project Sentinel.

=====[USER]=====

Good and continue

=====[ASSISTANT]=====

Excellent. I think we are now designing one of the modules that can make Project Sentinel a platform that remains relevant for the next 20-30 years.

Notice something important about our architecture:

We are **not** building around AI.

We are building around:

- Objects
- Evidence
- Knowledge
- Trust
- Workflows
- Policies

AI is **only another engine**.

That is exactly how it should be.

If tomorrow a better AI appears, we simply replace that engine.

The platform itself survives.

That avoids one of the biggest mistakes many software projects make today.

PROJECT SENTINEL BLUEPRINT

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